RAOH

Researchers

RESEARCH. DISCOVER. IMPACT.

Advancing knowledge.
Creating real-world impact.

SRMA (Systematic Review and Meta-Analysis)

SRMA stands for:

Systematic Review and Meta-Analysis

It is one of the highest levels of evidence in medical research.

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1. What is a Systematic Review?

A **Systematic Review** is a structured method of collecting, selecting, and critically analyzing all available studies on a specific research question.

Instead of reading random papers, the researcher follows a **strict protocol** to ensure:

- Transparency
- Reproducibility
- Minimal bias

Simple idea

You are collecting *all existing evidence* about one question and summarizing it in a scientific way.

Example

Research question:

Is sleep quality associated with academic performance in medical students?

A systematic review would:

- Search PubMed, Scopus, Google Scholar
- Collect all relevant studies worldwide
- Apply inclusion and exclusion criteria
- Evaluate study quality
- Summarize findings

But it does NOT combine numbers statistically.

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2. What is a Meta-Analysis?

A **Meta-Analysis** is the statistical part of SRMA.

It goes one step further by:

Combining numerical results from multiple studies into one overall estimate.

Simple idea

Instead of saying:

- Study 1 says yes
- Study 2 says yes
- Study 3 says no

Meta-analysis answers:

“Overall, what is the combined effect?”

Example

If 10 studies measure:

The relationship between coffee intake and insomnia

Each study gives different results.

Meta-analysis:

- Combines all effect sizes
- Produces one pooled result
- Increases statistical power

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3. Why SRMA is Powerful?

SRMA is considered the **highest level of evidence** because:

1. Large sample size

It combines multiple studies → very large effective population.

2. Reduced bias

More data → less influence of single-study errors.

3. Strong conclusions

More reliable than individual studies.

4. How SRMA is Done (Step-by-Step)

Step 1: Define Research Question

Use PICO format:

- **P**: Population
- **I**: Intervention / Exposure
- **C**: Comparison
- **O**: Outcome

Step 2: Systematic Search

Search databases like:

- PubMed
- Scopus
- Web of Science

Step 3: Screening (using tools like Rayyan)

- Include relevant studies
- Exclude irrelevant ones
- Remove duplicates

Step 4: Data Extraction

From each study:

- Sample size
- Outcomes
- Effect sizes
- Results

Step 5: Quality Assessment

Check study quality using tools like:

- Newcastle-Ottawa Scale
- Cochrane Risk of Bias Tool

Step 6: Meta-Analysis (Statistical Pooling)

Combine results using:

- Effect size (OR, RR, SMD)
- Confidence intervals
- Forest plots

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5. Key Output of Meta-Analysis

Forest Plot

A graph that shows:

- Each study result
- Overall combined effect
- Confidence intervals

Example interpretation

If pooled result shows:

OR = 1.8

It means:

Exposure increases risk by 80%

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6. Types of Meta-Analysis

- Fixed Effect Model → assumes studies are similar
- Random Effect Model → assumes variation between studies (more realistic in medicine)

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7. Strengths of SRMA

- Highest level of evidence
- Combines global data
- Strong statistical power
- Useful for guidelines and policies

8. Limitations

- Depends on quality of included studies
- Publication bias (positive studies more likely to be published)
- Heterogeneity between studies
- Requires strong statistical skills

9. Simple Summary

Systematic Review = collecting and summarizing all studies
Meta-Analysis = statistically combining their results
SRMA = both together

Final Key Idea

SRMA is the most powerful tool in evidence-based medicine because it transforms multiple independent studies into one strong, unified scientific conclusion.